



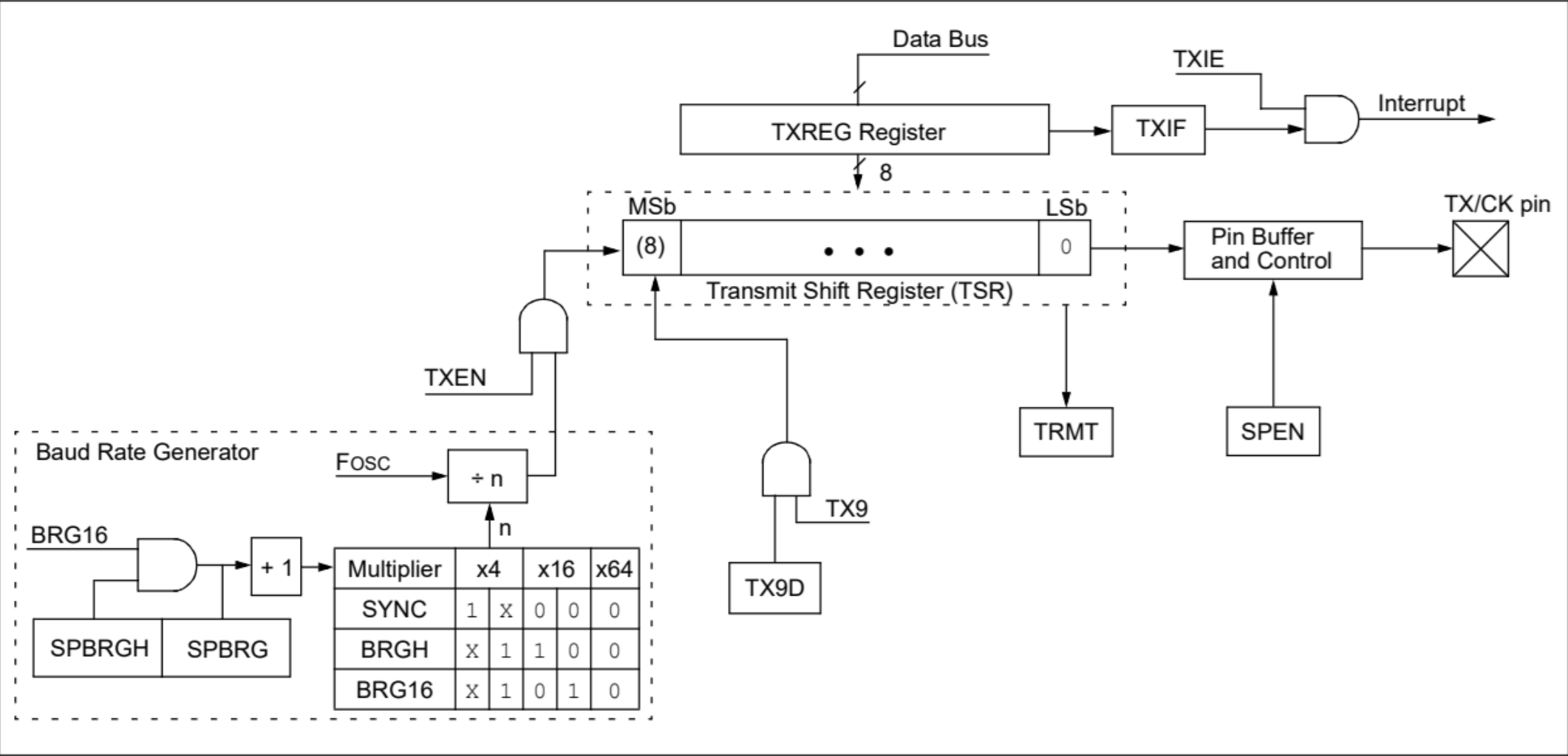
# ELECTRÓNICA DIGITAL 2

## CLASE 9 Módulo UART

Ing. Emiliano Migliore

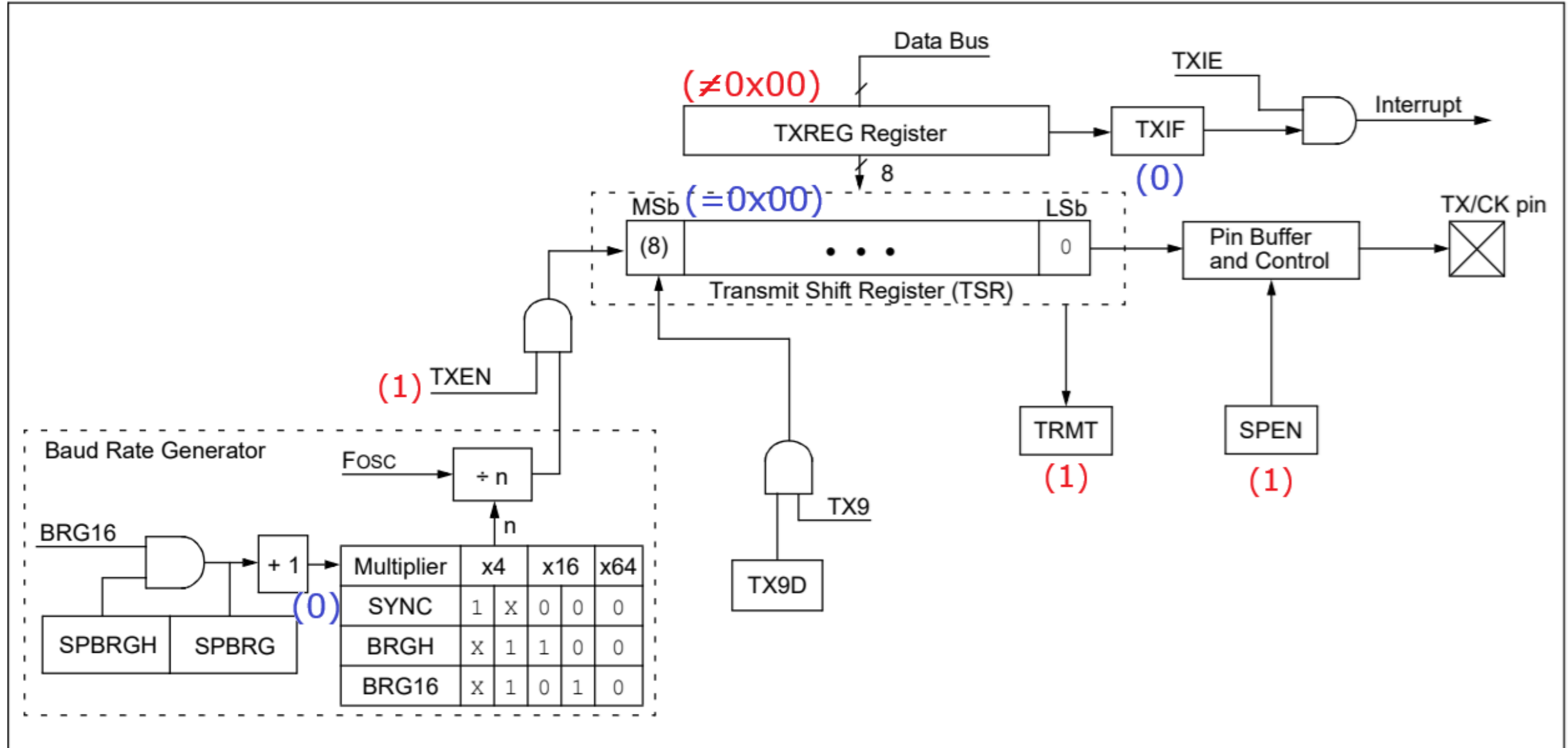
# Módulo EUSART: Transmisión Asíncrona (TX)

El módulo EUSART (Enhanced Universal Synchronous Asynchronous Receiver Transmitter) permite al MCU comunicarse con otros dispositivos a través de **comunicación serie**, ya sea en modo **síncrono** o **asíncrono**.



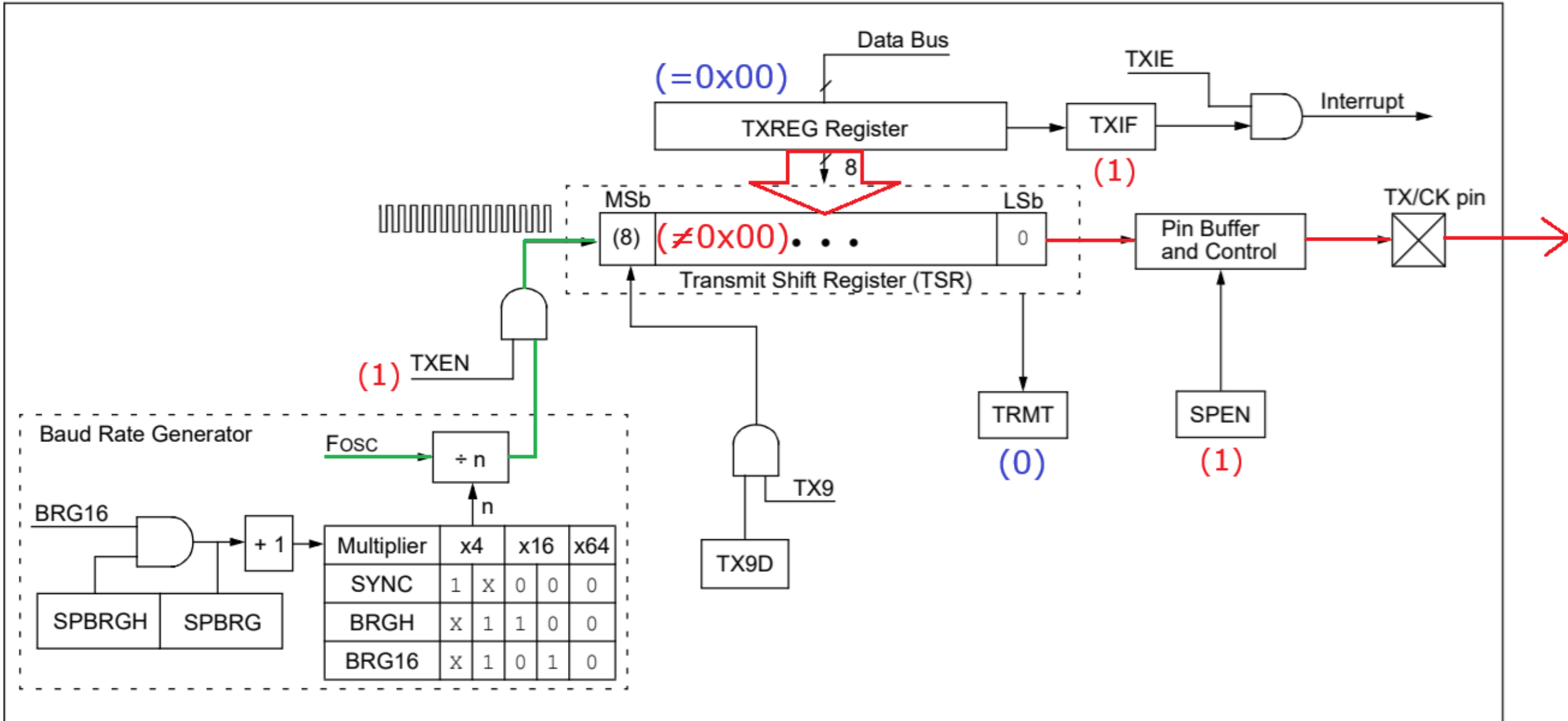
EUSART TRANSMIT BLOCK DIAGRAM

# Módulo EUSART: Transmisión Asíncrona (TX)



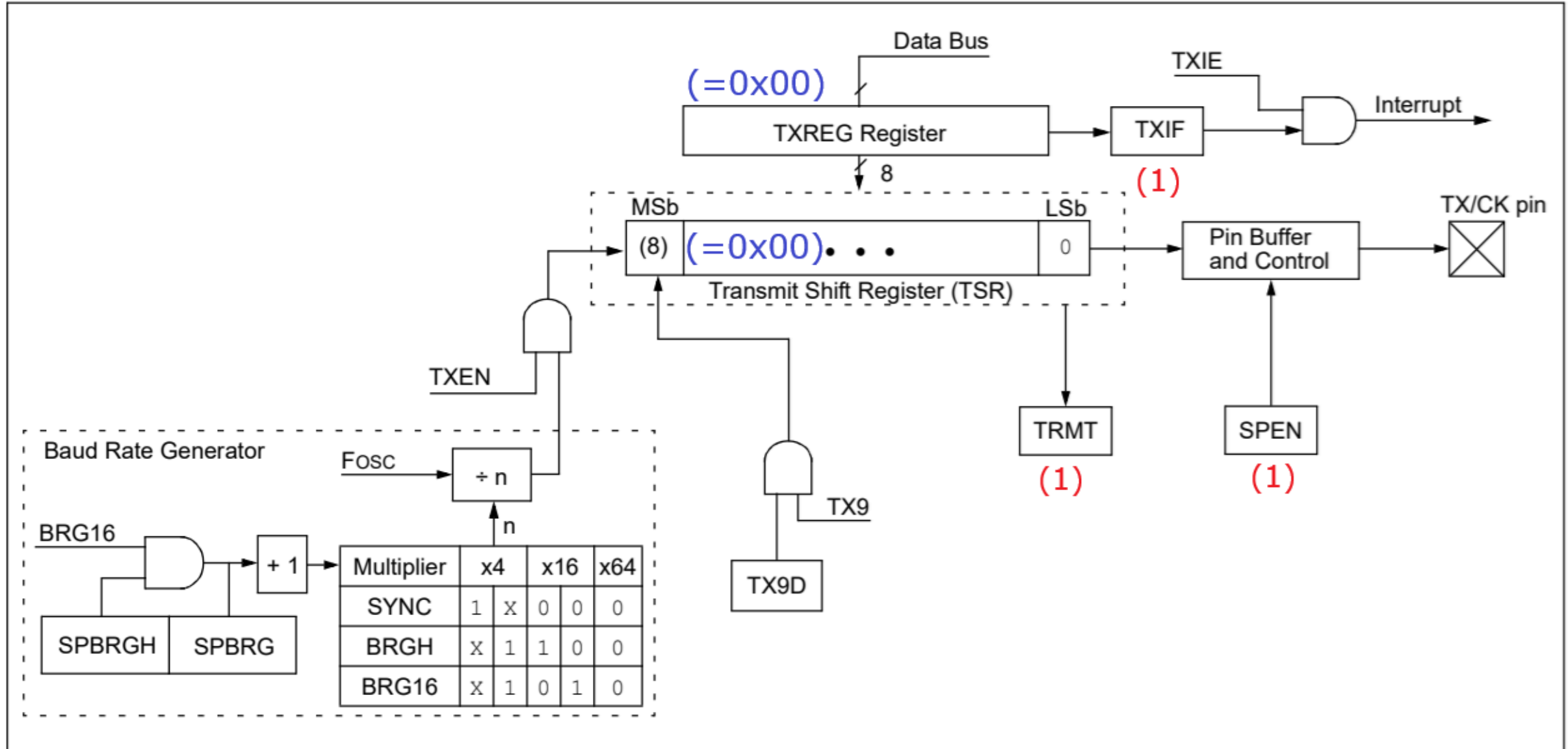
**EUSART TRANSMIT BLOCK DIAGRAM**  
(Antes de Transmitir)

## Módulo EUSART: Transmisión Asíncrona (TX)



### EUSART TRANSMIT BLOCK DIAGRAM (Durante la Transmisión)

# Módulo EUSART: Transmisión Asíncrona (TX)



**EUSART TRANSMIT BLOCK DIAGRAM**

(Luego de la Transmisión)

# Módulo EUSART: Transmisión Asíncrona (TX)

Para habilitar la transmisión, y configurar su modo se utilizan los siguientes bits:

## TXSTA: TRANSMIT STATUS AND CONTROL REGISTER

|       |       |                     |       |       |       |      |       |
|-------|-------|---------------------|-------|-------|-------|------|-------|
| R/W-0 | R/W-0 | R/W-0               | R/W-0 | R/W-0 | R/W-0 | R-1  | R/W-0 |
| CSRC  | TX9   | TXEN <sup>(1)</sup> | SYNC  | SENDB | BRGH  | TRMT | TX9D  |
| bit 7 |       |                     |       |       |       |      | bit 0 |

bit 5      **TXEN:** Transmit Enable bit<sup>(1)</sup>  
1 = Transmit enabled ➡ Tx ON  
0 = Transmit disabled

bit 1      **TRMT:** Transmit Shift Register Status bit  
1 = TSR empty  
0 = TSR full

bit 4      **SYNC:** EUSART Mode Select bit  
1 = Synchronous mode  
0 = Asynchronous mode ➡ Tx ASYNC

## RCSTA: RECEIVE STATUS AND CONTROL REGISTER

|       |       |       |       |       |      |      |       |
|-------|-------|-------|-------|-------|------|------|-------|
| R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R-0  | R-0  | R-x   |
| SPEN  | RX9   | SREN  | CREN  | ADDEN | FERR | OERR | RX9D  |
| bit 7 |       |       |       |       |      |      | bit 0 |

bit 7      **SPEN:** Serial Port Enable bit  
1 = Serial port enabled (configures RX/DT and TX/CK pins as serial port pins) ➡ RC6 Tx Mode  
0 = Serial port disabled (held in Reset)

# Módulo EUSART: Transmisión Asíncrona (TX)

Suponemos que la velocidad de transmisión en baudios deseada es de 9600 Bd:

## BAUD RATE FORMULAS

| Configuration Bits |       |      | BRG/EUSART Mode     | Baud Rate Formula    |
|--------------------|-------|------|---------------------|----------------------|
| SYNC               | BRG16 | BRGH |                     |                      |
| 0                  | 0     | 0    | 8-bit/Asynchronous  | $F_{osc}/[64 (n+1)]$ |
| 0                  | 0     | 1    | 8-bit/Asynchronous  | $F_{osc}/[16 (n+1)]$ |
| 0                  | 1     | 0    | 16-bit/Asynchronous |                      |
| 0                  | 1     | 1    | 16-bit/Asynchronous | $F_{osc}/[4 (n+1)]$  |
| 1                  | 0     | x    | 8-bit/Synchronous   |                      |
| 1                  | 1     | x    | 16-bit/Synchronous  |                      |

Legend: x = Don't care, n = value of SPBRGH, SPBRG register pair

## REGISTERS ASSOCIATED WITH THE BAUD RATE GENERATOR

| Name   | Bit 7 | Bit 6 | Bit 5 | Bit 4 | Bit 3 | Bit 2 | Bit 1 | Bit 0 |
|--------|-------|-------|-------|-------|-------|-------|-------|-------|
| SPBRG  | BRG7  | BRG6  | BRG5  | BRG4  | BRG3  | BRG2  | BRG1  | BRG0  |
| SPBRGH | BRG15 | BRG14 | BRG13 | BRG12 | BRG11 | BRG10 | BRG9  | BRG8  |

## BAUD RATES FOR ASYNCHRONOUS MODES

| BAUD RATE | SYNC = 0, BRGH = 1, BRG16 = 0 |         |                       |                   |         |                       |                  |         |                       |                  |         |                       |
|-----------|-------------------------------|---------|-----------------------|-------------------|---------|-----------------------|------------------|---------|-----------------------|------------------|---------|-----------------------|
|           | Fosc = 4.000 MHz              |         |                       | Fosc = 3.6864 MHz |         |                       | Fosc = 2.000 MHz |         |                       | Fosc = 1.000 MHz |         |                       |
|           | Actual Rate                   | % Error | SPBRG value (decimal) | Actual Rate       | % Error | SPBRG value (decimal) | Actual Rate      | % Error | SPBRG value (decimal) | Actual Rate      | % Error | SPBRG value (decimal) |
| 300       | —                             | —       | —                     | —                 | —       | —                     | —                | —       | —                     | 300              | 0.16    | 207                   |
| 1200      | 1202                          | 0.16    | 207                   | 1200              | 0.00    | 191                   | 1202             | 0.16    | 103                   | 1202             | 0.16    | 51                    |
| 2400      | 2404                          | 0.16    | 103                   | 2400              | 0.00    | 95                    | 2404             | 0.16    | 51                    | 2404             | 0.16    | 25                    |
| 9600      | 9615                          | 0.16    | 25                    | 9600              | 0.00    | 23                    | 9615             | 0.16    | 12                    | —                | —       | —                     |
| 10417     | 10417                         | 0.00    | 23                    | 10473             | 0.53    | 21                    | 10417            | 0.00    | 11                    | 10417            | 0.00    | 5                     |
| 19.2k     | 19.23k                        | 0.16    | 12                    | 19.2k             | 0.00    | 11                    | —                | —       | —                     | —                | —       | —                     |
| 57.6k     | —                             | —       | —                     | 57.60k            | 0.00    | 3                     | —                | —       | —                     | —                | —       | —                     |
| 115.2k    | —                             | —       | —                     | 115.2k            | 0.00    | 1                     | —                | —       | —                     | —                | —       | —                     |

$$Bd_{Deseado} = \frac{f_{CLK}}{16(n + 1)} = \frac{f_{CLK}}{16(SPBRG + 1)} \Rightarrow SPBRG = \frac{f_{CLK}}{16Bd} - 1$$

$$SPBRG = \frac{4x10^6Hz}{16 * 9600} - 1 = 25,042 \Rightarrow SPBRG = 25$$

$$Bd_{Calculado} = \frac{4x10^6Hz}{16(25 + 1)} = 9615,38 \Rightarrow Bd_{Calculado} = 9615$$

$$\varepsilon_{Bd} [\%] = \frac{Bd_{Calculado} - Bd_{Deseado}}{Bd_{Deseado}}$$

$$\varepsilon_{Bd} = \frac{9615 - 9600}{9600} * 100 = 0,156\%$$

# Módulo EUSART: Transmisión Asíncrona (TX)

## BAUD RATE FORMULAS

| Configuration Bits |       |      | BRG/EUSART Mode     | Baud Rate Formula    |
|--------------------|-------|------|---------------------|----------------------|
| SYNC               | BRG16 | BRGH |                     |                      |
| 0                  | 0     | 0    | 8-bit/Asynchronous  | $F_{osc}/[64 (n+1)]$ |
| 0                  | 0     | 1    | 8-bit/Asynchronous  | $F_{osc}/[16 (n+1)]$ |
| 0                  | 1     | 0    | 16-bit/Asynchronous |                      |
| 0                  | 1     | 1    | 16-bit/Asynchronous | $F_{osc}/[4 (n+1)]$  |
| 1                  | 0     | x    | 8-bit/Synchronous   |                      |
| 1                  | 1     | x    | 16-bit/Synchronous  |                      |

## BAUDCTL: BAUD RATE CONTROL REGISTER

|        |       |     |       |       |     |       |       |
|--------|-------|-----|-------|-------|-----|-------|-------|
| R-0    | R-1   | U-0 | R/W-0 | R/W-0 | U-0 | R/W-0 | R/W-0 |
| ABDOVF | RCIDL | —   | SCKP  | BRG16 | —   | WUE   | ABDEN |
| bit 7  |       |     |       |       |     |       | bit 0 |

bit 3      **BRG16:** 16-bit Baud Rate Generator bit  
1 = 16-bit Baud Rate Generator is used  
0 = 8-bit Baud Rate Generator is used

## TXSTA: TRANSMIT STATUS AND CONTROL REGISTER

|       |       |                     |       |       |       |      |       |
|-------|-------|---------------------|-------|-------|-------|------|-------|
| R/W-0 | R/W-0 | R/W-0               | R/W-0 | R/W-0 | R/W-0 | R-1  | R/W-0 |
| CSRC  | TX9   | TXEN <sup>(1)</sup> | SYNC  | SENDB | BRGH  | TRMT | TX9D  |
| bit 7 |       |                     |       |       |       |      | bit 0 |

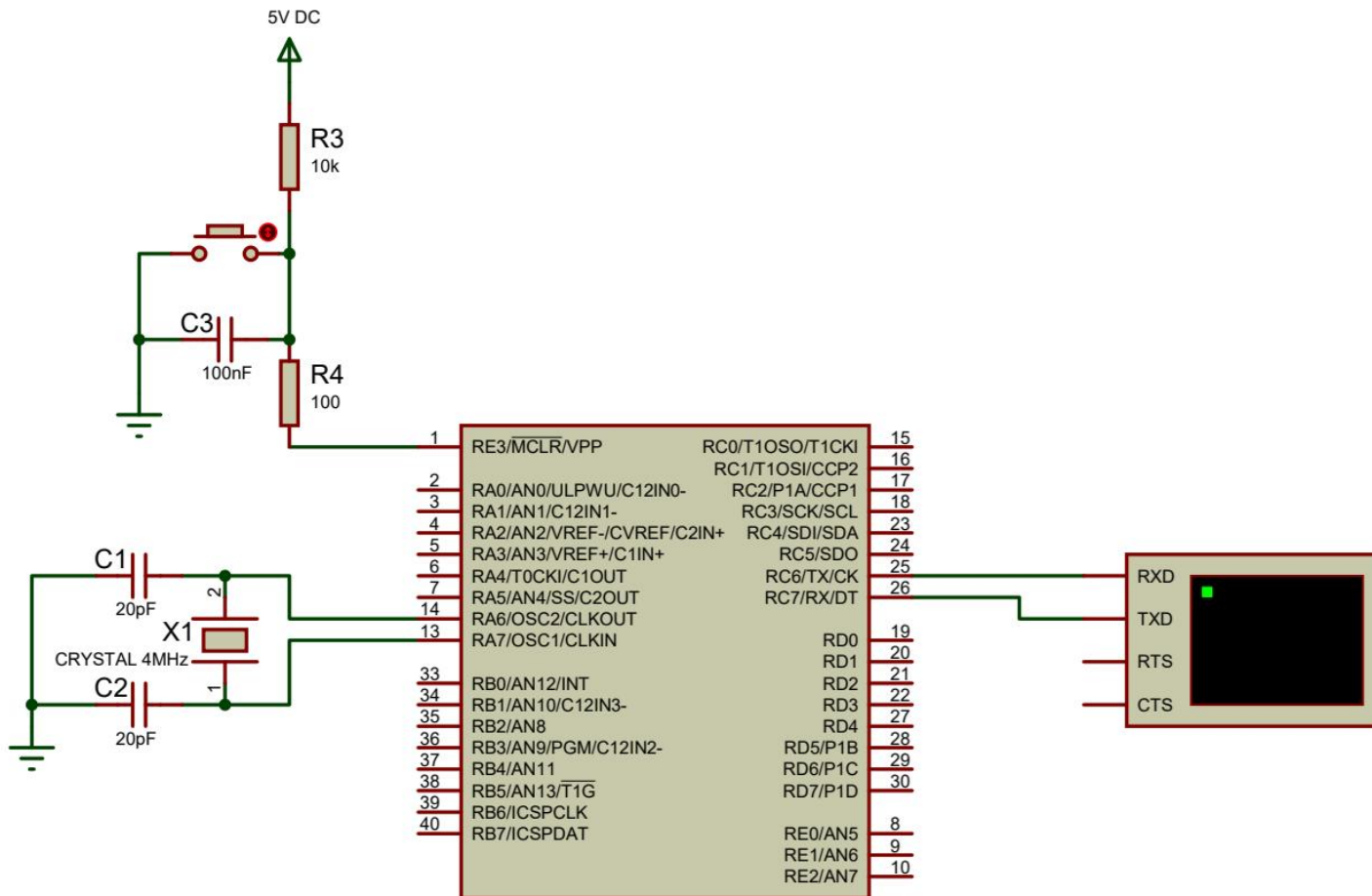
bit 4      **SYNC:** EUSART Mode Select bit  
1 = Synchronous mode  
0 = Asynchronous mode

bit 2      **BRGH:** High Baud Rate Select bit  
Asynchronous mode:  
1 = High speed  
0 = Low speed



# Ejercicio Anexo 6: Circuito de Comunicación de Datos

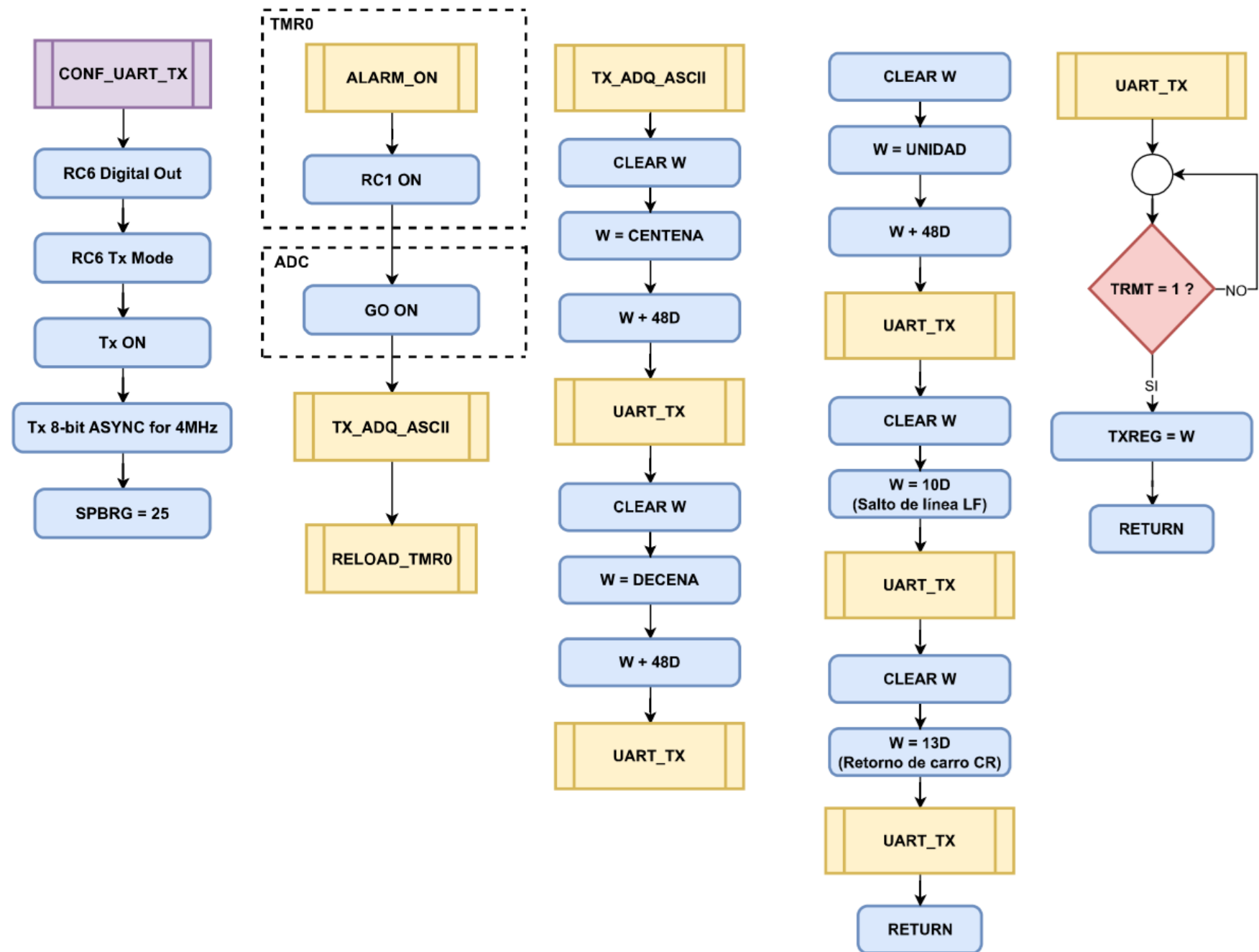
El siguiente ejemplo consiste en la implementación de un algoritmo en ASM que sea capaz de realizar la transmisión y recepción de datos a partir de comunicación serie, mediante el periférico EUSART del PIC16F887.



USB to UART TTL Module



# Diseño del Software: Diagramas de Flujo



| ASCII Hex Símbolo |   |     | ASCII Hex Símbolo |    |   |
|-------------------|---|-----|-------------------|----|---|
| 0                 | 0 | NUL | 48                | 30 | 0 |
| 1                 | 1 | SOH | 49                | 31 | 1 |
| 2                 | 2 | STX | 50                | 32 | 2 |
| 3                 | 3 | ETX | 51                | 33 | 3 |
| 4                 | 4 | EOT | 52                | 34 | 4 |
| 5                 | 5 | ENQ | 53                | 35 | 5 |
| 6                 | 6 | ACK | 54                | 36 | 6 |
| 7                 | 7 | BEL | 55                | 37 | 7 |
| 8                 | 8 | BS  | 56                | 38 | 8 |
| 9                 | 9 | TAB | 57                | 39 | 9 |
| 10                | A | LF  | 58                | 3A | : |
| 11                | B | VT  | 59                | 3B | ; |
| 12                | C | FF  | 60                | 3C | < |
| 13                | D | CR  | 61                | 3D | = |
| 14                | E | SO  | 62                | 3E | > |
| 15                | F | SI  | 63                | 3F | ? |